Complex Numbers: Cartesian Form & Algebra

ANSWER KEY

Section 1: Cartesian Arithmetic & Complex Conjugate

1 Given $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$, calculate $z_1 z_2$.

a. $5 + 14i$ ✓

b. $-3 - 8i$

c. $5 - 14i$

d. $-3 + 14i$

$z_1 z_2 = (3 - 2i)(-1 + 4i) = -3 + 12i + 2i - 8i^2 = -3 + 14i + 8 = 5 + 14i.$

2 Express $\frac{4 + 2i}{1 - i}$ in Cartesian form $a + bi$.

a. $1 + 3i$ ✓

b. $3 + 3i$

c. $1 - 3i$

d. $2 + 2i$

Multiply numerator and denominator by conjugate $(1 + i)$: $\frac{(4 + 2i)(1 + i)}{(1 - i)(1 + i)} = \frac{4 + 4i + 2i + 2i^2}{1 - i^2} = \frac{2 + 6i}{2} = 1 + 3i.$

3 If $z = x + iy$, what is $z \cdot z^*$ (where z^* is the complex conjugate)?

a. $x^2 + y^2$ ✓

b. $x^2 - y^2$

c. $x^2 + 2ixy - y^2$

d. $2x$

$zz^* = (x + iy)(x - iy) = x^2 - (iy)^2 = x^2 - i^2 y^2 = x^2 + y^2 = |z|^2.$

Section 2: Modulus and Argument

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Find the modulus $|z|$ and principal argument $\text{Arg}(z) \in (-\pi, \pi]$ for $z = -\sqrt{3} + i$.

a. $|z| = 2, \text{Arg}(z) = \frac{5\pi}{6}$ ✓

b. $|z| = 2, \text{Arg}(z) = -\frac{\pi}{6}$

c. $|z| = 4, \text{Arg}(z) = \frac{2\pi}{3}$

d. $|z| = 2, \text{Arg}(z) = \frac{7\pi}{6}$

$|z| = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = 2$. In Quadrant II ($x < 0, y > 0$), reference angle is $\arctan(1/\sqrt{3}) = \frac{\pi}{6}$. Thus $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

5 What is the modulus of $\frac{3+4i}{1-2i}$?

a. $\sqrt{5}$ ✓

b. $\frac{5}{3}$

c. 5

d. $\frac{25}{5} = 5$

Using modulus properties: $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|} = \frac{\sqrt{3^2+4^2}}{\sqrt{1^2+(-2)^2}} = \frac{5}{\sqrt{5}} = \sqrt{5}$.

6 Find the principal argument of $z = -5i$.

a. $-\frac{\pi}{2}$ ✓

b. $\frac{3\pi}{2}$

c. $\frac{\pi}{2}$

d. π

The point lies on the negative imaginary axis, which corresponds to angle $-\frac{\pi}{2}$ within the principal range $(-\pi, \pi]$.

Section 3: Solving Complex Equations

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Solve the equation $2z + 3z^* = 15 - 2i$ for $z = x + iy$.

a. $z = 3 + 2i$ ✓

b. $z = 3 - 2i$

c. $z = 5 - 2i$

d. $z = 2 + 3i$

$2(x + iy) + 3(x - iy) = 5x - iy = 15 - 2i$. Equating real parts: $5x = 15 \Rightarrow x = 3$. Imaginary parts: $-y = -2 \Rightarrow y = 2$. Thus $z = 3 + 2i$.

8 Find the two square roots of $-8 - 6i$ in Cartesian form.

a. $\pm(1 - 3i)$ ✓

b. $\pm(3 - i)$

c. $\pm(1 + 3i)$

d. $\pm(2 - 3i)$

Let $(x + iy)^2 = x^2 - y^2 + 2ixy = -8 - 6i$. Thus $x^2 - y^2 = -8$ and $2xy = -6 \Rightarrow y = -3/x$. $x^2 - 9/x^2 = -8 \Rightarrow x^4 + 8x^2 - 9 = 0 \Rightarrow (x^2 + 9)(x^2 - 1) = 0$. Since $x \in \mathbb{R}$, $x = \pm 1$, giving $y = \mp 3$. Roots are $\pm(1 - 3i)$.

9 Solve the quadratic equation $z^2 - 4z + 13 = 0$ over $\mathbb{C}$.

a. $z = 2 \pm 3i$ ✓

b. $z = 4 \pm 6i$

c. $z = -2 \pm 3i$

d. $z = 2 \pm 9i$

By quadratic formula: $z = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$.

Section 4: Argand Diagram Geometry & Loci

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Describe the locus of points in the complex plane satisfying $|z - 3 + 2i| = 5$.

a. **A circle with center** $(3, -2)$ **and radius** 5 ✓

b. A circle with center $(-3, 2)$ and radius 5

c. A straight line perpendicular to $3 - 2i$

d. An ellipse with focal distance 5

Rewrite as $|z - (3 - 2i)| = 5$. The distance from z to the fixed point $3 - 2i$ is constant and equal to 5, which defines a circle centered at $(3, -2)$ with radius 5.

11 What geometric locus is defined by $|z - 1| = |z - 3i|$?

a. **The perpendicular bisector of the segment connecting** $(1, 0)$ **and** $(0, 3)$ ✓

b. A circle passing through $(1, 0)$ and $(0, 3)$

c. A line segment connecting $(1, 0)$ and $(0, 3)$

d. A parabola opening upward

$|z - z_1| = |z - z_2|$ represents the set of all points equidistant from $z_1 = 1 + 0i$ and $z_2 = 0 + 3i$, which is the perpendicular bisector of the segment connecting them.

12 Find the Cartesian equation of the locus $\text{Arg}(z - 2) = \frac{\pi}{4}$.

a. **The half-line** $y = x - 2$ **for** $x > 2$ ✓

b. The full line $y = x - 2$

c. The circle $(x - 2)^2 + y^2 = 1$

d. The half-line $y = -x + 2$ for $x < 2$

$\text{Arg}((x - 2) + iy) = \frac{\pi}{4} \Rightarrow \frac{y}{x-2} = \tan(\pi/4) = 1 \Rightarrow y = x - 2$, with the ray restriction $x - 2 > 0 \Rightarrow x > 2, y > 0$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647